Population-based Multimodal Continuous Optimization

Assignment — Theoretical Base

MetaLab Suite

Problem #3

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Abstract

A swarm of sensors must triangulate the **strongest rescue signal** over a continuous, "n"-dimensional terrain riddled with **false positives** (local optima). The terrain is a classic multimodal benchmark function, shifted so that its single global optimum — the real signal — sits at a random location. The scarce resource is *not time*: it is the **evaluation budget**. Every call to the terrain consumes one evaluation; the environment counts them and cuts (or penalises) any call beyond the limit. This document defines the problem; the technical instructions for running the environment, the black-box solver interface and the autograder are in the companion guide `multimodal_continuous_optimization/README.md`.

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1 Assignment brief

The sensors can query the terrain at any point and read a scalar value: the **terrain cost**. They never see the mathematical function, its name, its gradient or the location of the signal — only the value they measure. The goal is to report, as accurately as possible, the **coordinates of the strongest signal** (the global optimum), using **at most** a fixed number of evaluations.

You must implement **one population-based metaheuristic** (Differential Evolution, Particle Swarm Optimization, a Genetic Algorithm, CMA-ES, Artificial Bee Colony, . . .) that operates through the black-box callback alone. The only algorithm supplied is a plain random search.

2 The search space

The domain is the axis-aligned box

$$\mathcal{X} = [L, U]^n \subset \mathbb{R}^n, \qquad L < U, \tag{1}$$

where the dimension n and the bounds (L, U) depend on the instance (e.g. $n = 2$ with $[L, U] = [-5.12, 5.12]$ for the *toy* terrain, $n = 10$ with $[-500, 500]$ for the *hard* one). A *candidate* is a point $x \in \mathcal{X}$; the swarm must decide which points to query.

3 The terrain: a family of classic multimodal functions

The terrain is one member $f : \mathbb{R}^n \to \mathbb{R}$ of a family of **classic multimodal benchmark functions**. Each has a single global minimum surrounded by many local minima — the false positives that lure a myopic search. Three representatives are

$$f_{\text{ras}}(x) = 10n + \sum_{i=1}^{n} \Big(x_i^2 - 10\cos(2\pi x_i) \Big), \qquad \text{(Rastrigin)} \tag{2}$$

$$f_{\text{ack}}(x) = -20\exp\Big(-0.2\sqrt{\tfrac{1}{n}\textstyle\sum_i x_i^2}\Big) - \exp\Big(\tfrac{1}{n}\textstyle\sum_i \cos(2\pi x_i)\Big) + 20 + e, \qquad \text{(Ackley)} \tag{3}$$

$$f_{\text{sch}}(x) = 418.9829\, n - \sum_{i=1}^{n} x_i \sin\big(\sqrt{|x_i|}\big). \qquad \text{(Schwefel)} \tag{4}$$

Table 1 lists the family the instances are drawn from, with the canonical bounds and the *canonical* optimum x^{c} (the argmin in the function's own frame) and its value $f(x^{\text{c}})$.

Table 1: The multimodal family used to build terrains.

Function	Domain	Canonical argmin x^{c}	$f(x^{\text{c}})$
Rastrigin	$[-5.12, 5.12]$	0	0
Ackley	$[-32.768, 32.768]$	0	0
Griewank	$[-600, 600]$	0	0
Schwefel	$[-500, 500]$	420.9687	0
Levy	$[-10, 10]$	1	0
Styblinski–Tang	$[-5, 5]$	-2.9035	$-39.166\, n$
Drop-Wave (2-D)	$[-5.12, 5.12]$	0	-1
Schaffer N.2 (2-D)	$[-100, 100]$	0	0

4 The hidden signal: shifting the terrain

To hide the answer, the terrain is *shifted* so that its global optimum lands at a random location $x^\star$ (the **signal**). Given the canonical argmin x^{c}, the environment evaluates

$$g(x) = f(x - x^\star + x^{\text{c}}), \qquad \text{so that} \qquad g(x^\star) = f(x^{\text{c}}) = \min_{x \in \mathcal{X}} g(x), \tag{5}$$

where $x^\star$ is drawn uniformly from the interior of the box (kept a margin away from the boundary). A candidate is scored by g, and the coordinates the swarm reports are compared against the hidden $x^\star$.

Remark 1 (Domain-limited functions)**.** *A few functions are bounded below only on their canonical box — most notably* ***Schwefel****, which diverges to $-\infty$ outside $[-500, 500]$. Such functions are* not *shifted ($x^\star = x^{\text{c}}$); otherwise the "global optimum" would drift to the boundary. The generator enforces this, so every instance is well posed: $g(x^\star)$ is always the true minimum.*

5 The objective

"Strongest signal" means *lowest terrain cost*: the environment returns $g(x)$, and the swarm must **minimise** it. Equivalently, the signal strength is $-g(x)$ and the false positives are the local minima of g. The quality of a run is measured by the **distance** between the reported coordinates and the hidden signal,

$$\text{error} = \frac{\|\hat{x} - x^\star\|}{\sqrt{n}\,(U - L)} \in [0, 1], \tag{6}$$

normalised so that 0 is a perfect hit and 1 is the worst point in the box.

6 The constraint: the evaluation budget

Unlike a simulation-time problem, there is *no clock* inside the model. The single scarce resource is the number of evaluations

$$\boxed{\sum_k \mathbf{1}\{\text{the swarm queries the terrain}\} \;\le\; B} \tag{7}$$

for an instance-dependent budget B (e.g. $B = 2\,000$ for *toy*, $B = 100\,000$ for *hard*). The environment is the referee:

- every call to the terrain **counts** one evaluation;
- once the budget is spent, a further call is **cut** (the environment raises and the run is aborted) or **penalised** (a large constant is returned), depending on the mode;
- either way the run is flagged **invalid**, so exceeding the budget can never beat an honest run.

Remark 2 (Why this is the hard part)**.** *A large budget rewards methods that* use it well*, not those that iterate most: a population that balances exploration (to leave a false positive) and exploitation (to refine the signal) will triangulate the optimum long before a random sampler or a trajectory method would.*

7 What you are given

The following are provided (see the companion guide for the exact commands):

- the **environment engine** (`given/simulator_mmco.py`), which builds the terrain, exposes the black-box evaluation and enforces the budget;
- the **instance generator** (`given/generate_instances_mmco.py`), which builds the `toy` and `hard` terrains and guarantees they are well posed;
- a **black-box student API** (`student_mmco.py`) whose only window onto the terrain is the callback `evaluate(coords)`;
- the **only algorithm supplied**, a random search (`random_search_mmco.py`);
- a **terminal test harness** (`run_local.py`) and the **autograder backend** (`autograder/`).

8 Deliverables

1. A population-based metaheuristic that finds the signal **within the budget**, using only the black-box callback.
2. Results for the `toy` and `hard` terrains: best coordinates, the error (6), the evaluations used and the convergence curve.
3. A short discussion of the exploration/exploitation balance under a fixed evaluation budget.

References

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